Linear Regression

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What is Linear Regression?

Goal: Learn a straight-line relationship between input and output.

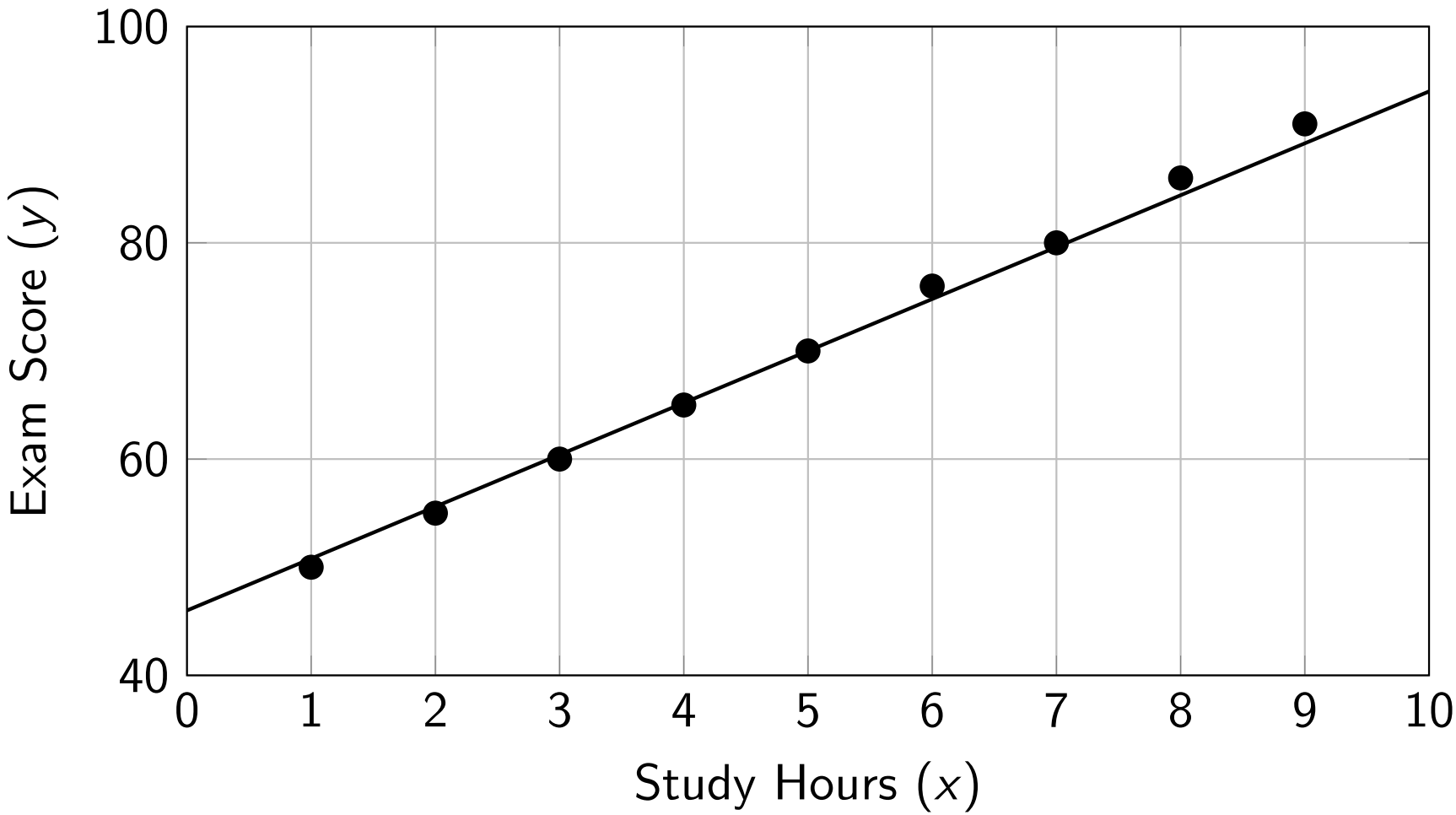
$$\hat{y} = wx + b$$

- x = input feature
- $\hat{y}$ = predicted output
- w = slope of the line
- b = intercept

Interpretation:

- As x changes, $\hat{y}$ changes linearly.
- Linear regression tries to find the **best-fitting line**.

Visual Example: Best-Fit Line



Example: More study hours generally lead to higher exam scores.

How Does the Model Make Predictions?

Suppose the learned line is:

$$\hat{y} = 4.8x + 46$$

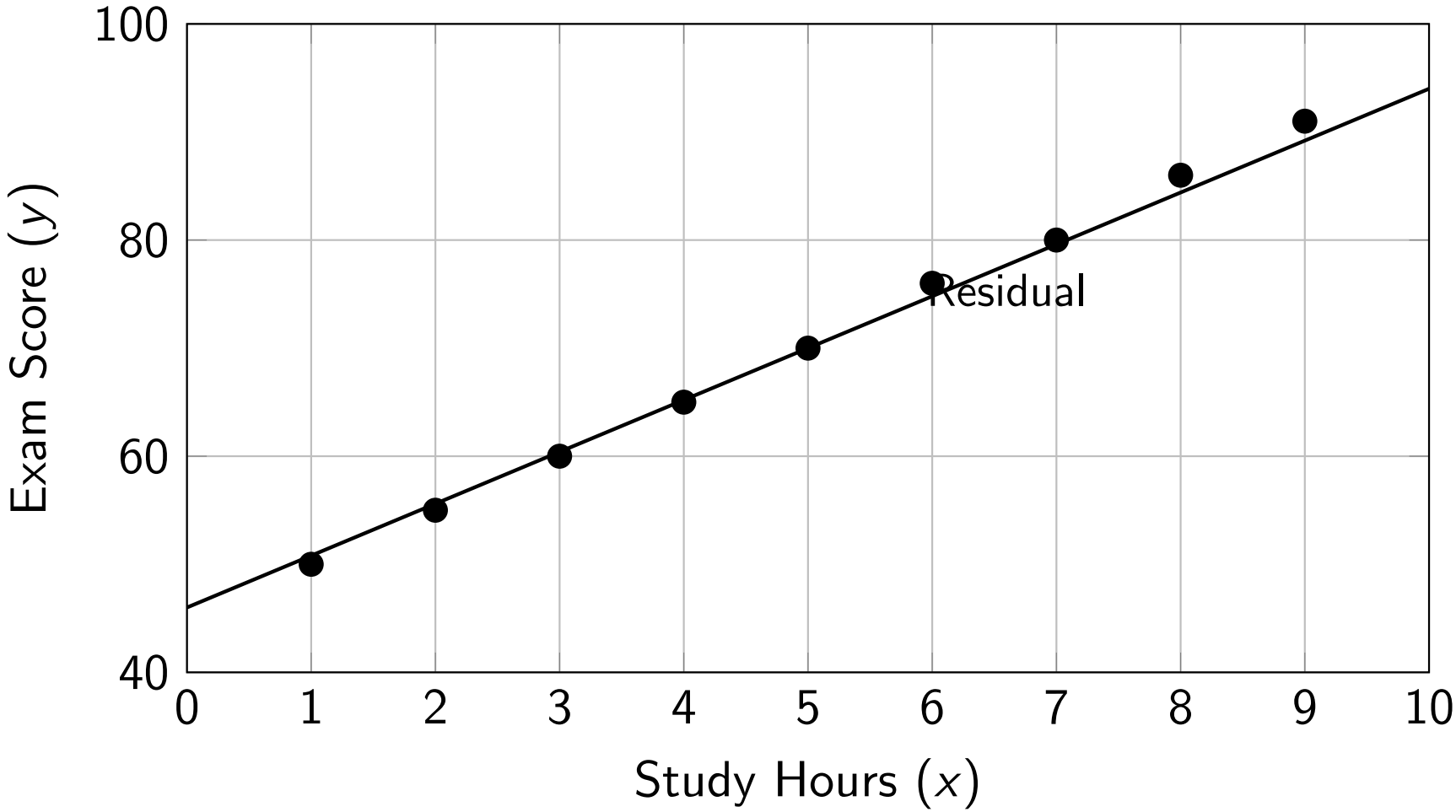
If a student studies for $x = 5$ hours, then:

$$\hat{y} = 4.8(5) + 46 = 70$$

Meaning:

- The model predicts an exam score of 70.
- The line allows us to estimate outputs for new inputs.

Residual Error: Distance from the Line



Residual:

$$\text{Residual} = y - \hat{y}$$

It measures how far a data point is from the regression line

Training Objective

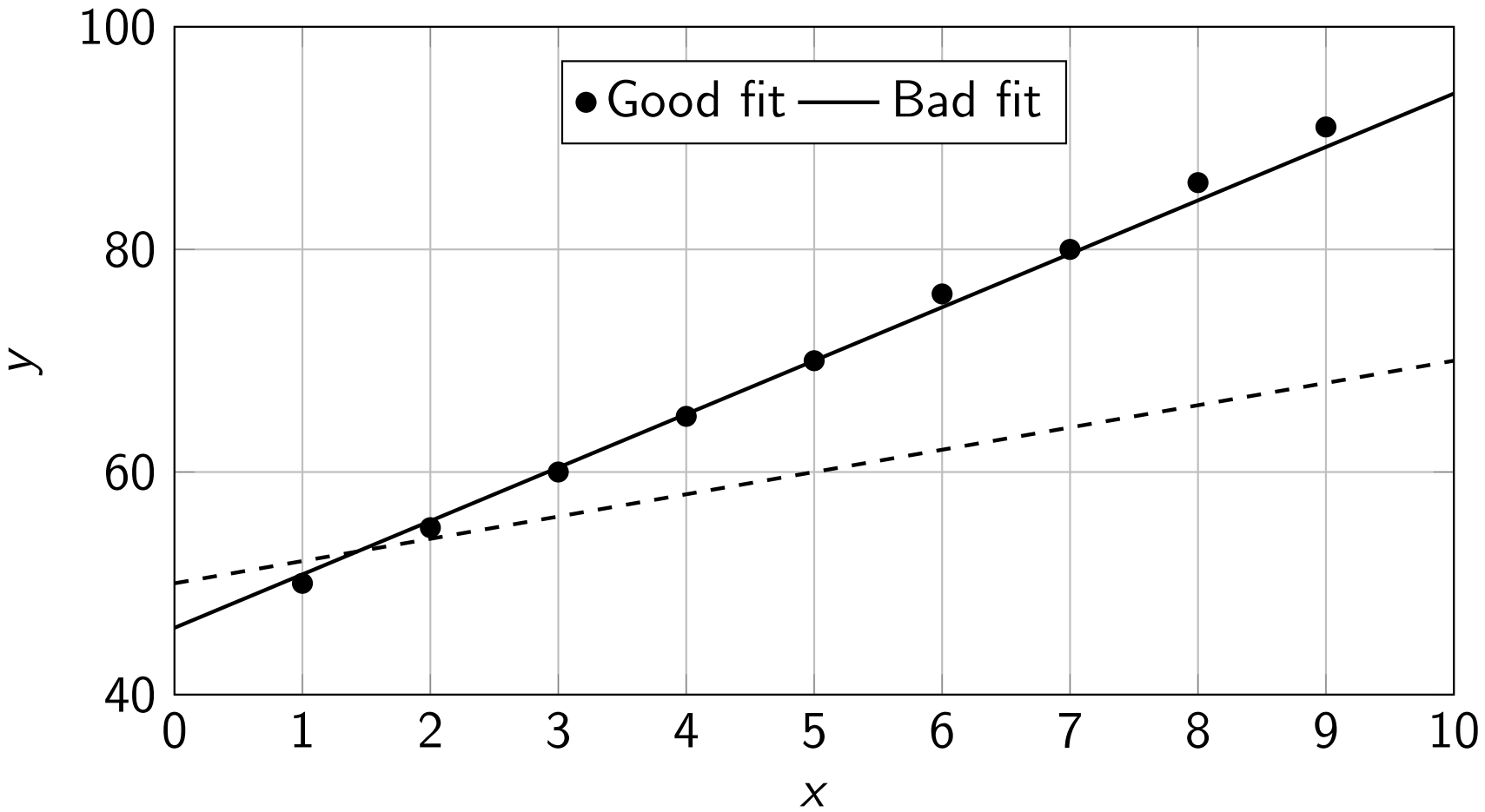
Linear regression chooses w and b such that prediction errors are as small as possible.

Common loss function: Mean Squared Error (MSE)

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

- y_i = true value
- $\hat{y}_i$ = predicted value
- Square is used so that large errors are penalized more

Good Fit vs Bad Fit



A good regression line stays closer to the data points overall.

When Does Linear Regression Work Well?

Works well when:

- Relationship between variables is approximately linear
- Noise is not too large
- Trend can be captured by a straight line

Examples:

- Study hours vs exam score
- Advertising budget vs sales
- House size vs price

Limitations of Linear Regression

Problem: Not all relationships are linear.

- Cannot model curves well
- Sensitive to outliers
- Assumes a simple straight-line pattern

This is why AI moves beyond linear regression to:

- Polynomial models
- Logistic regression
- Neural networks

Loss Function: Measuring Error

Goal:

How do we quantify how wrong our model is?

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$$e_i = y_i - \hat{y}_i$$

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Mean Squared Error (MSE):

$$\mathcal{L}(w, b) = \frac{1}{n} \sum_{i=1}^{n} (y_i - (wx_i + b))^2$$

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Why square the error?

- Penalizes large errors more
- Always positive
- Smooth and differentiable (important for optimization)

Learning = Optimization Problem

Objective:

$$(w^*, b^*) = \arg\min_{w,b} \mathcal{L}(w, b)$$

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Key Insight:

Training a model = solving an optimization problem

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Interpretation:

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Key Insight:

Training a model = solving an optimization problem

Question:

- How do we find the minimum?

Gradient Descent: How Learning Happens

Idea:

Move parameters in the direction that reduces loss

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Move parameters in the direction that reduces loss

Update rule:

$$w = w - \eta \frac{\partial \mathcal{L}}{\partial w}$$

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Where:

- η = learning rate (step size)
- Gradient = direction of steepest increase

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Where:

- $\eta =$ learning rate (step size)
- Gradient $=$ direction of steepest increase

So we move in:

$$-\nabla \mathcal{L} \quad \text{(steepest decrease)}$$

Gradient for Linear Regression

Loss:

$$\mathcal{L}(w, b) = \frac{1}{n} \sum (y_i - (wx_i + b))^2$$

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Gradient w.r.t b:

$$\frac{\partial \mathcal{L}}{\partial b} = -\frac{2}{n} \sum (y_i - \hat{y}_i)$$

Insight:

- Error $\times$ input determines update

Gradient Descent: Intuition

Think of a landscape:

- Loss function = height
- Parameters (w, b) = position

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Process:

- Start somewhere
- Look at slope
- Move downhill
- Repeat until convergence

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Key idea:

- Gradient tells us which direction to move

From Single Input to Multiple Inputs

Model:

$$\hat{y}_i = w_1 x_{i1} + w_2 x_{i2} + \cdots + w_d x_{id} + b$$

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where

$$\mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_d \end{bmatrix}, \qquad \mathbf{x}_i = \begin{bmatrix} x_{i1} \\ x_{i2} \\ \vdots \\ x_{id} \end{bmatrix}$$

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Loss function:

$$\mathcal{L}(\mathbf{w}, b) = \frac{1}{n} \sum_{i=1}^{n} \Big(y_i - (\mathbf{w}^T \mathbf{x}_i + b) \Big)^2$$

Gradient Derivation w.r.t. One Weight

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To compute the gradient for one parameter w_j:

$$\frac{\partial \mathcal{L}}{\partial w_j} = \frac{\partial}{\partial w_j} \left[\frac{1}{n} \sum_{i=1}^{n} \Big(y_i - (\mathbf{w}^T \mathbf{x}_i + b) \Big)^2 \right]$$

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Apply the chain rule:

$$= \frac{1}{n} \sum_{i=1}^{n} 2 \left(y_i - \hat{y}_i \right) \frac{\partial}{\partial w_j} \Big(y_i - (\mathbf{w}^T \mathbf{x}_i + b) \Big)$$

Finishing the Derivation

Recall:

$$\mathbf{w}^T \mathbf{x}_i = \sum_{k=1}^{d} w_k x_{ik}$$

So,

$$\frac{\partial}{\partial w_j} \Big(y_i - (\mathbf{w}^T \mathbf{x}_i + b) \Big) = -x_{ij}$$

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Interpretation:

Gradient Derivation w.r.t. Bias

$$\mathcal{L}(\mathbf{w}, b) = \frac{1}{n} \sum_{i=1}^{n} \left(y_i - (\mathbf{w}^T \mathbf{x}_i + b) \right)^2$$

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$$\mathcal{L}(\mathbf{w}, b) = \frac{1}{n} \sum_{i=1}^{n} \left(y_i - (\mathbf{w}^T \mathbf{x}_i + b) \right)^2$$

Differentiate with respect to b:

$$\frac{\partial \mathcal{L}}{\partial b} = \frac{1}{n} \sum_{i=1}^{n} \frac{\partial}{\partial b} \left(y_i - (\mathbf{w}^T \mathbf{x}_i + b) \right)^2$$

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Apply the chain rule:

$$= \frac{1}{n}\sum_{i=1}^{n} 2(y_i - \hat{y}_i)\frac{\partial}{\partial b}\left(y_i - (\mathbf{w}^T\mathbf{x}_i + b)\right)$$

Since

$$\frac{\partial}{\partial b}\left(y_i - (\mathbf{w}^T\mathbf{x}_i + b)\right) = -1$$

we get

$$\boxed{\frac{\partial \mathcal{L}}{\partial b} = -\frac{2}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)}$$

Vector Form of the Gradient

Instead of deriving each weight separately, we can write the gradient compactly.

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Gradient w.r.t. bias:

$$\boxed{\frac{\partial \mathcal{L}}{\partial b} = -\frac{2}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)}$$

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Update rules:

$$\mathbf{w} \leftarrow \mathbf{w} - \eta \nabla_{\mathbf{w}} \mathcal{L}$$

$$b \leftarrow b - \eta \frac{\partial \mathcal{L}}{\partial b}$$

Matrix Form (Advanced View)

Let

$$X \in \mathbb{R}^{n \times d}, \qquad \mathbf{y} \in \mathbb{R}^{n}, \qquad \hat{\mathbf{y}} = X\mathbf{w} + b\mathbf{1}$$

Matrix Form (Advanced View)

Let

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The gradients become

$$\boxed{\nabla_{\mathbf{w}}\mathcal{L} = -\frac{2}{n}X^T(\mathbf{y} - \hat{\mathbf{y}})}$$

$$\boxed{\frac{\partial \mathcal{L}}{\partial b} = -\frac{2}{n}\mathbf{1}^T(\mathbf{y} - \hat{\mathbf{y}})}$$

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Why this matters:

- This same idea extends directly to neural networks
- Backpropagation is repeated chain rule on larger compositions

Takeaway

- Linear regression models a relationship using a straight line
- Prediction function:

$$\hat{y} = wx + b$$

- The best line minimizes prediction error
- It is simple, interpretable, and foundational for machine learning

Worked Example: Gradient Descent with Multiple Features

Consider the following data points:

X_1	X_2	Y_t
1	2	40
2	3	90

We use the hypothesis:

$$Y_p = w_1 x_1^4 + w_2 x_2^3 + b$$

and the Mean Squared Error loss:

$$\mathcal{L}(w_1, w_2, b) = \frac{1}{n} \sum_{i=1}^{n} (y_{t,i} - y_{p,i})^2$$

Question

Using the data and hypothesis below:

$$Y_p = w_1 x_1^4 + w_2 x_2^3 + b$$

X_1	X_2	Y_t
1	2	40
2	3	90

Answer the following:

1. Calculate the loss magnitude using Mean Squared Error. [3]
2. If $w_1 = 0.3$, $w_2 = 0.4$, and $b = 0.7$, apply one step of gradient descent to update w_2. Use learning rate $\eta = 0.0001$. [5]
3. If Mean Squared Error is replaced by Mean Cubic Error, what potential issues may arise? [2]

Solution Step 1: Compute Predictions

Given:

$$w_1 = 0.3, \quad w_2 = 0.4, \quad b = 0.7$$

Hypothesis:

$$Y_p = w_1 x_1^4 + w_2 x_2^3 + b$$

For data point 1: $(x_1, x_2, y_t) = (1, 2, 40)$

$$y_{p,1} = 0.3(1^4) + 0.4(2^3) + 0.7$$

$$y_{p,1} = 0.3 + 0.4(8) + 0.7 = 0.3 + 3.2 + 0.7 = 4.2$$

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$$y_{p,1} = 0.3 + 0.4(8) + 0.7 = 0.3 + 3.2 + 0.7 = 4.2$$

For data point 2: $(x_1, x_2, y_t) = (2, 3, 90)$

$$y_{p,2} = 0.3(2^4) + 0.4(3^3) + 0.7$$

$$y_{p,2} = 0.3(16) + 0.4(27) + 0.7 = 4.8 + 10.8 + 0.7 = 16.3$$

Solution Step 2: Compute MSE Loss

Mean Squared Error:

$$\mathcal{L} = \frac{1}{n}\sum_{i=1}^{n}(y_{t,i} - y_{p,i})^2$$

Here $n = 2$, so

$$\mathcal{L} = \frac{1}{2}\left[(40 - 4.2)^2 + (90 - 16.3)^2\right]$$

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Compute each term:

$$40 - 4.2 = 35.8, \qquad (35.8)^2 = 1281.64$$

$$90 - 16.3 = 73.7, \qquad (73.7)^2 = 5431.69$$

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$$40 - 4.2 = 35.8, \qquad (35.8)^2 = 1281.64$$

$$90 - 16.3 = 73.7, \qquad (73.7)^2 = 5431.69$$

Therefore,

$$\mathcal{L} = \frac{1}{2}(1281.64 + 5431.69)$$

$$\mathcal{L} = \frac{6713.33}{2} = 3356.665$$

Solution Step 3: Derive Gradient for w_2

Loss:

$$\mathcal{L} = \frac{1}{n}\sum_{i=1}^{n}(y_{t,i} - y_{p,i})^2$$

where

$$y_{p,i} = w_1 x_{1,i}^4 + w_2 x_{2,i}^3 + b$$

For w_2,

$$\frac{\partial \mathcal{L}}{\partial w_2} = -\frac{2}{n}\sum_{i=1}^{n} x_{2,i}^3 (y_{t,i} - y_{p,i})$$

Since $n = 2$,

$$\frac{\partial \mathcal{L}}{\partial w_2} = -\sum_{i=1}^{2} x_{2,i}^3 (y_{t,i} - y_{p,i})$$

Solution Step 4: Update w_2

We already have:

$$y_{p,1} = 4.2, \qquad y_{p,2} = 16.3$$

Errors:

$$y_{t,1} - y_{p,1} = 40 - 4.2 = 35.8$$

$$y_{t,2} - y_{p,2} = 90 - 16.3 = 73.7$$

Also,

$$x_{2,1}^3 = 2^3 = 8, \qquad x_{2,2}^3 = 3^3 = 27$$

Solution Step 4: Update w_2

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$$y_{t,2} - y_{p,2} = 90 - 16.3 = 73.7$$

Also,

$$x_{2,1}^3 = 2^3 = 8, \qquad x_{2,2}^3 = 3^3 = 27$$

So,

$$\frac{\partial \mathcal{L}}{\partial w_2} = -\left[8(35.8) + 27(73.7)\right]$$

$$= -\left[286.4 + 1989.9\right] = -2276.3$$

Solution Step 4: Update w_2

We already have:

$$y_{p,1} = 4.2, \qquad y_{p,2} = 16.3$$

Errors:

$$y_{t,1} - y_{p,1} = 40 - 4.2 = 35.8$$

$$y_{t,2} - y_{p,2} = 90 - 16.3 = 73.7$$

Also,

$$x_{2,1}^3 = 2^3 = 8, \qquad x_{2,2}^3 = 3^3 = 27$$

So,

$$\frac{\partial \mathcal{L}}{\partial w_2} = -\left[8(35.8) + 27(73.7)\right]$$

$$= -\left[286.4 + 1989.9\right] = -2276.3$$

continue

Gradient descent update:

$$w_2^{\text{new}} = w_2 - \eta \frac{\partial \mathcal{L}}{\partial w_2}$$

$$w_2^{\text{new}} = 0.4 - 0.0001(-2276.3)$$

$$w_2^{\text{new}} = 0.4 + 0.22763 = 0.62763$$

$$\boxed{w_2^{\text{new}} = 0.62763}$$

Solution Step 5: If MSE is Replaced by Mean Cubic Error

Suppose the loss becomes:

$$\mathcal{L}_{\text{cubic}} = \frac{1}{n}\sum_{i=1}^{n}(y_{t,i} - y_{p,i})^3$$

Potential issues:

- **Negative and positive errors may cancel out**, so the loss may not reflect total error properly.
- **Optimization becomes unstable**, because cubic terms can produce very large gradients for large errors.
- **Loss is not always non-negative**, unlike MSE, which makes interpretation harder.

MSE is usually preferred because it is non-negative, smooth, and stable.